Content Broadcasting API docs

1 Global API Conventions

To ensure consistency across all client integrations, this API adheres to strict response envelopes and standard HTTP status codes.

1.1 Standard Response Envelope

Every response (both successful and failed) is wrapped in a consistent JSON envelope.

```
{
  "success": boolean,
  "message": "string (Human readable description)",
  "data": object | array | null,
  "stack": "string (Development environment only, omitted in Prod)"
}
```

1.2 Standard HTTP Status Codes

- **200 OK** - Request succeeded.
- **201 Created** - Resource successfully created (Upload/Register).
- **400 Bad Request** - Validation failed (missing fields, bad formats).
- **401 Unauthorized** - Missing, invalid, or expired JWT.
- **403 Forbidden** - Valid JWT, but insufficient role permissions (RBAC).
- **413 Payload Too Large** - File exceeds strict byte limits.
- **500 Internal Server Error** - Unhandled exception (caught by Global Handler).

2 Content Lifecycle & Role Workflow

The broadcasting engine operates on a strict state machine. Content will **never** be broadcast unless it reaches the `approved` state and is within its active time window.

The State Machine Lifecycle

1. **Teacher Uploads:** Status defaults to `pending`. (Excluded from Broadcast).
2. **Principal Reviews:**
   - **Approve:** Status changes to `approved`. → *Eligible for Broadcast.*
   - **Reject:** Status changes to `rejected`. Reason is recorded. → *Excluded.*

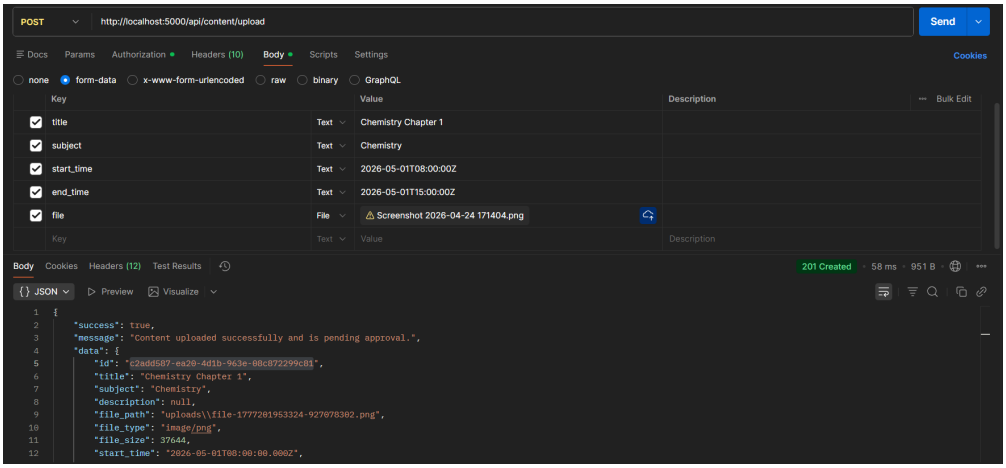

Figure 1: API response for `POST /content/upload` request (captured via Postman)

	status character varying (20)	rejection_reason text	uploaded_by uuid	approved_by uuid
1	rejected	Image is too blurry, please upload a higher resolution version.	a59cd4f0-72bb-46d9-9272-f53de319a43d	58cc0821-4fde-40c7-bb8a-f7944a83b201
2	approved	[null]	a59cd4f0-72bb-46d9-9272-f53de319a43d	58cc0821-4fde-40c7-bb8a-f7944a83b201
3	approved	[null]	a59cd4f0-72bb-46d9-9272-f53de319a43d	58cc0821-4fde-40c7-bb8a-f7944a83b201

Figure 2: Database entries showing review-related fields: `status`, `rejection_reason`, `uploaded_by`, and `approved_by`

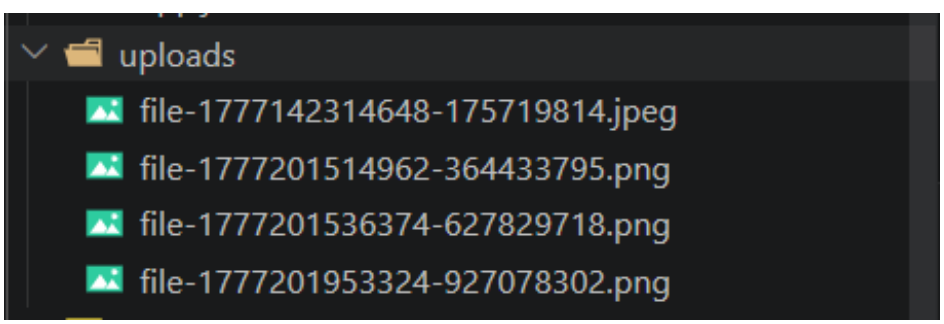

Figure 3: Uploaded files appearing in the `uploads/` directory after a successful `POST /content/upload` request

3 Authentication (RBAC)

3.1 Register User

POST `/api/auth/register`
Validation Rules:

- `email`: Must match valid RFC 5322 regex.
- `password`: Minimum 6 characters.
- `role`: Strictly limited to enum `['teacher', 'principal']`.

Request Payload Schema

```
{
  "name": "string (Required, max 100 chars)",
  "email": "string (Required, valid email format)",
  "password": "string (Required, min 6 chars)",
  "role": "string (Required, 'teacher' | 'principal')"
}
```

201 Created (Success)

```
{
  "success": true,
  "message": "User registered successfully",
  "data": { "id": "uuid", "name": "Abhishek", "role": "teacher" }
}
```

3.2 Login User

POST `/api/auth/login`
Rate Limit: Max 10 attempts per IP per hour.

401 Unauthorized (Failure)

```
{
  "success": false,
  "message": "Invalid email or password.",
  "data": null
}
```

4 Content Management

4.1 Upload Content

POST `/api/content/upload`
Auth: Bearer Token required. Role: ***`teacher`*** *only.*

Validation Rules:

- **File Size:** Hard limit of 10MB (10485760 bytes). Drops connection on exceed.
- **MIME Type:** `image/jpeg`, `image/png`, `image/gif`. (Extension spoofing rejected).
- **Time bounds:** `start_time` must be strictly $<$ `end_time`.
- **Duration:** `rotation_duration` must be integer > 0. Defaults to 5 if omitted.

400 Bad Request (Validation Failure)

```
{
  "success": false,
  "message": "Start time must be before end time.",
  "data": null
}
```

413 Payload Too Large (File Failure)

```
{
  "success": false,
  "message": "File too large. Maximum size is 10MB.",
  "data": null
}
```

4.2 Review Content

PATCH `/api/content/:id/review`
Auth: Bearer Token required. Role: ***`principal`*** *only.*

Request Payload Schema

```
{
  "status": "string (Required, 'approved' | 'rejected')",
  "rejection_reason": "string (Required IF status === 'rejected')"
}
```

403 Forbidden (RBAC Failure)

```
{
  "success": false,
  "message": "User role 'teacher' is not authorized to access this route.",
  "data": null
}
```

5 Broadcasting Engine

5.1 Get Live Rotation

GET `/api/broadcast/live/:teacherId`
Auth: Public (None required).

Behavior: Calculates the active frame using $(t_{elapsed} \pmod{D_{total}})$. If the teacher has no approved content, or content is outside the time window, the API fails gracefully with a 200 OK (not 404) to prevent client screen crashes.

200 OK (Empty State / No Content)

```
{
  "success": true,
  "message": "No active content to broadcast at this time.",
  "data": null
}
```

200 OK (Active Rotation Hit)

```
{
  "success": true,
  "message": "Live content retrieved",
  "data": {
    "id": "dc4d1db4-bece-45a9-bb18-fc7ab0548126",
    "title": "Math Homework Page 1",
    "subject": "Mathematics",
    "file_path": "uploads/file-1777142314648.jpeg",
    "rotation_duration": 5
  }
}
```